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Basic Heatmap Explanation and Generation

This article contains information about heatmaps within the iMotions platform.

This guide contains the following topics:

1. Basic Heatmap Explanation
2. Basic Heatmap Generation (within iMotions)

Basic Heatmap Explanation

Heatmaps, as a form of eye tracking (ET) data visualization and analysis, can be generated within iMotions in order to look at the areas of interest of a stimuli. The example below uses an image as the stimuli.

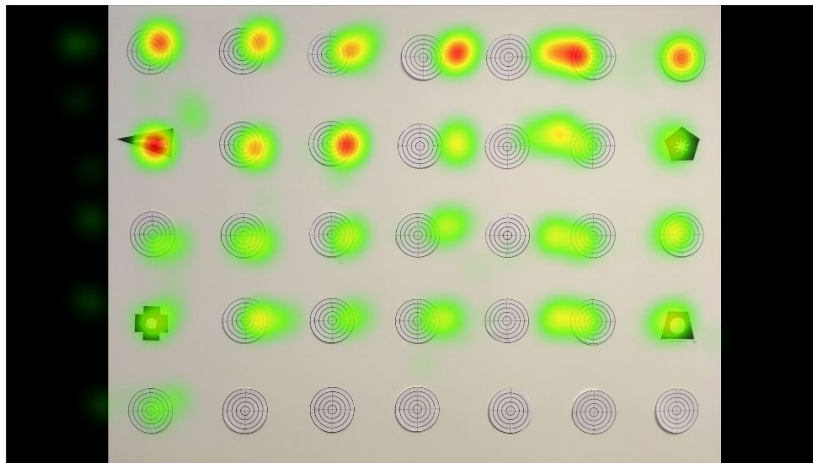
You can think of a heatmap as a data “paint by numbers” overlaid on top of an image.



The ET values are a complex calculation that includes the number of participants and how much time is spent per pixel. These values are calculated as “points” and are assigned to each pixel. The pixel with the greatest number of points represents the high end of the scale and is represented by the dark red color.



The heat scale can be thought of as quartiles – red indicates relatively more attention and green is relatively less attention.

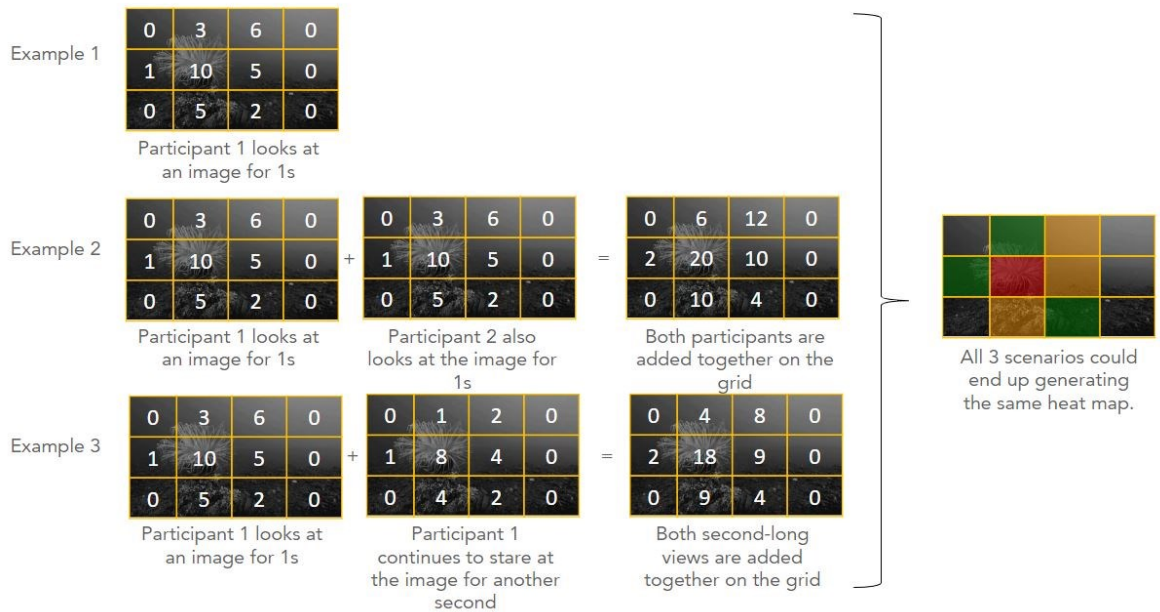


This heat can help identify what areas of the content was focused on for the longest amount of time, and what areas or elements may go missed entirely.

In the example above....

- The top row of targets was more focused on for a longer period of time, as indicated by the red color.
- The next few targets are primarily green and yellow in color, indicating that they were not focused on for as long of a time as the first few targets.
- Finally, the last few targets have no color overlay, indicating they were not looked at all.

The above example was generated from a single participant. However, there are multiple scenarios in which a heatmap can be created:



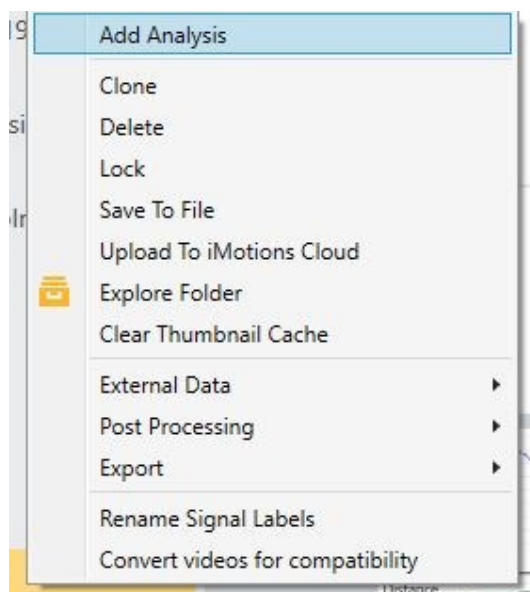
Basic Heatmap Generation

As for an explanation of how a heat map analysis can be generated, this will be shown below:

After recording all of your desired number of participants, you can begin a data analysis from the eye tracking data:

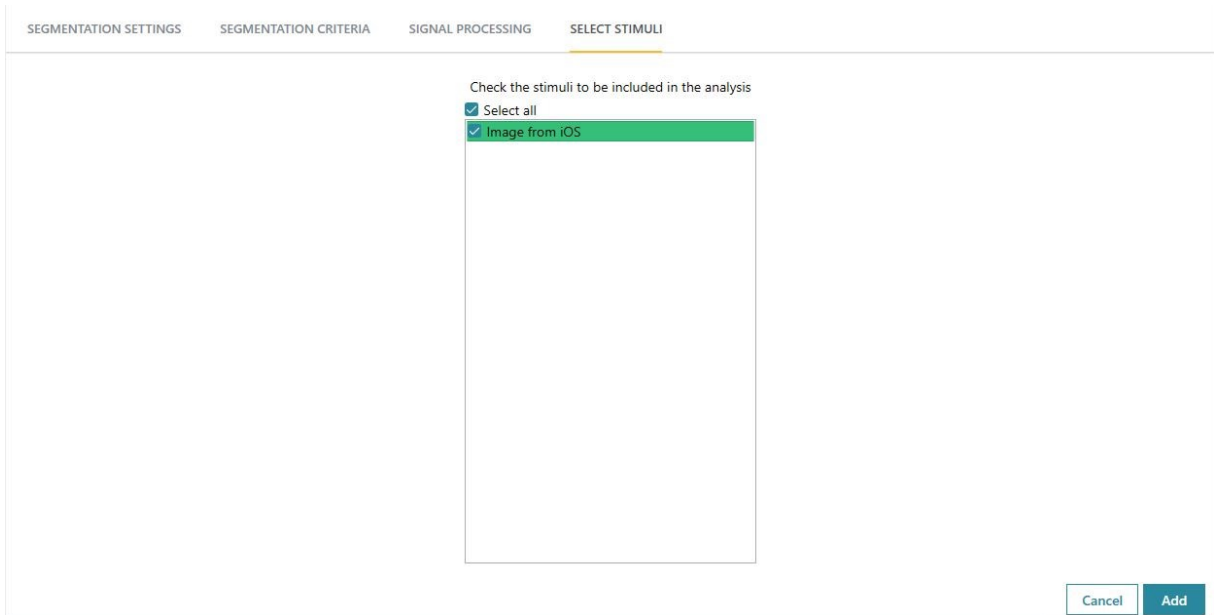
Step 1

- Right click on on your chosen study under “*Library*” on the left side on the iMotions platform, and select “*Add Analysis*” at the top of the menu:



Step 2

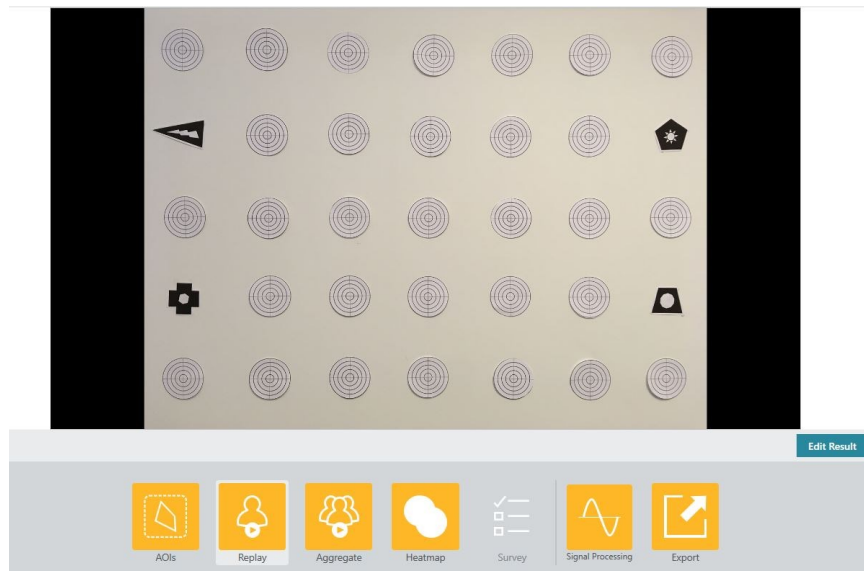
- The next screen to appear is labeled as “*Segmentation Settings*”, and you can either add your chosen settings, criteria, and signal processing, or click next through the pages to the last tab of “*Select Stimuli*” in order to generate a heatmap.
 - From here, you can select which stimuli you want to generate a heatmap for. In this example, it is an image. Click “*Add*”:



The screenshot shows the 'SELECT STIMULI' tab of a software interface. At the top, there are four tabs: 'SEGMENTATION SETTINGS', 'SEGMENTATION CRITERIA', 'SIGNAL PROCESSING', and 'SELECT STIMULI'. The 'SELECT STIMULI' tab is active and highlighted with a yellow underline. Below the tabs, there is a large white area. In the center of this area, there is a smaller white box. Inside this box, at the top, is the text 'Check the stimuli to be included in the analysis'. Below this text are two checkboxes. The first checkbox is labeled 'Select all' and is checked. The second checkbox is labeled 'Image from iOS' and is also checked. The 'Image from iOS' checkbox and its label are highlighted with a green background. At the bottom right of the main white area, there are two buttons: 'Cancel' and 'Add'.

Step 3

- What will appear next is an image of the stimuli, with a menu at the bottom of it. This will allow you to choose what analysis you want to run, and this is where you can select “*Heatmap*”:



Step 4

- Step 4: After loading (this will take longer with more participants/data), the heatmap will be generated:

